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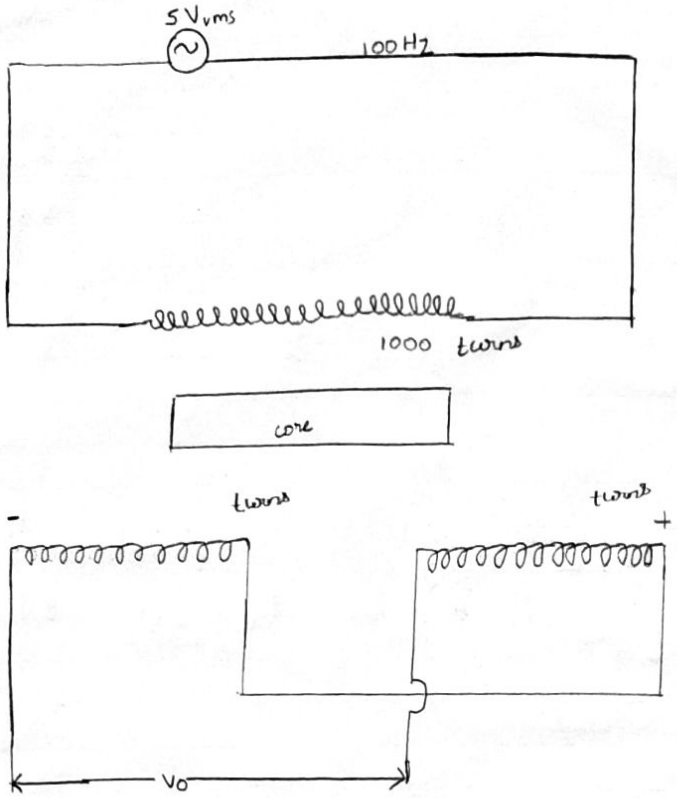
Circuit Diagram



Tabulation

Input Voltage		Output Voltage	frequency	Time = $\frac{1}{f}$
Positive Displacement	7	2.2V D = 5 mm	1000	$\frac{1}{1000}$ = 1mS
Negative Displacement	7	D = 5mm	1000	1mS

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Characteristic of LVDT
(characteristic variable differental Transformer)

Aim:

To understand and ~~st~~ simulate the relation between core displacement and output voltage

Apparatus Required

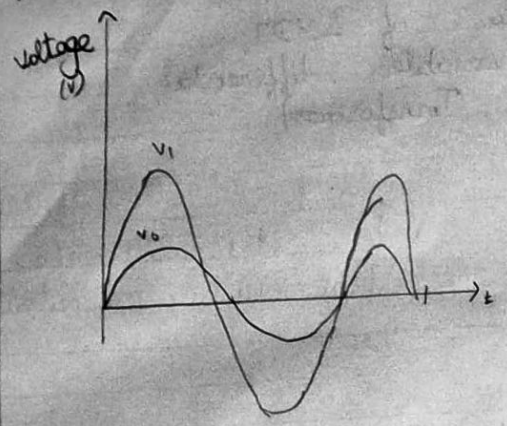
Laptop with internet connection

Theory

LVDT is an inductive transture that converts linear displacement into a dectrical signal. It consist of a transformer having one primary, winding and two secondary winding wound on a core. As the primary is connected to an AC source, the AC current and the voltage are produced in the secondary of LVDT. when the core is at the center, the flux linking with both the secondary winding are equal when the core is at the right, the flux linking with the seconda-ry winding S_1 is more

Model Graph

positive Displacement



calculation

$V_m = \sqrt{2} \times V_{rms}$

$= \sqrt{2} \times 5$

$= 7V$

$T = 1ms$

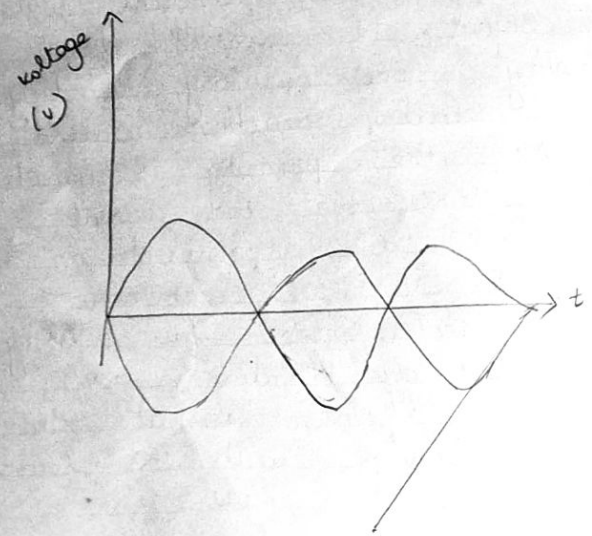
4 SD = 7

$SD = \frac{7}{4} V$

$1.25 SD = \frac{7}{4} \times 1.25$

$= 2.2V$

Negative Displacement



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So, the emf induced in S_1 is more than S_2. So, the net emf is positive when the core is at the left, the flux linking and the emf in S_2 is more than S_1. So, the net emf is negative

Procedure

* Connections are made as per the circuit diagram

* Set, the number of turns supplied voltage, and supplied frequency

* Configure the parameters

* Move the core to positive side and the negative side and observe the input voltage and output voltage wave forms

* Plot the graph between input and output voltage

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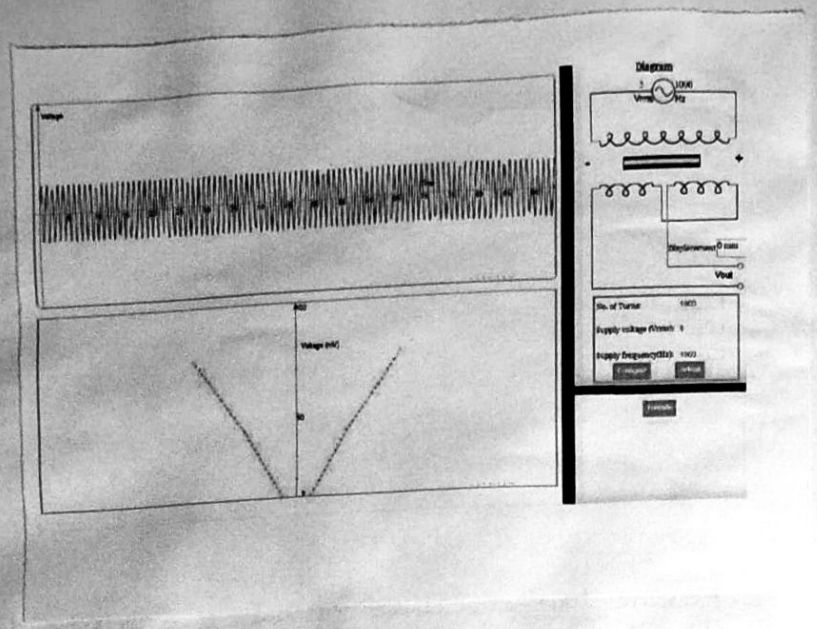

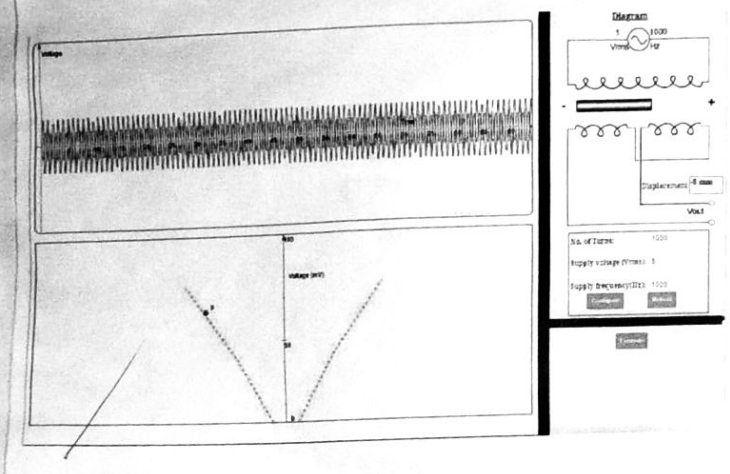

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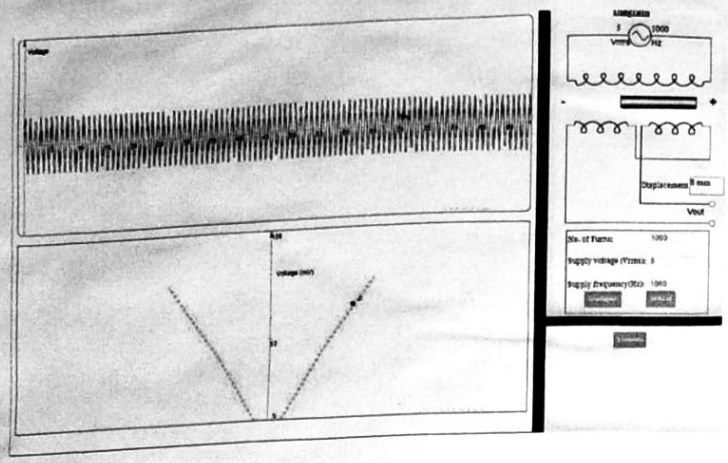

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Result

To understand and simulate the relation between core displacement and output voltage were developed and tested successfully